Lecture 9: Support Vector Machines

STATS 202: Data Mining and Analysis

Linh Tran

stat202@gmail.com

Department of Statistics
Stanford University

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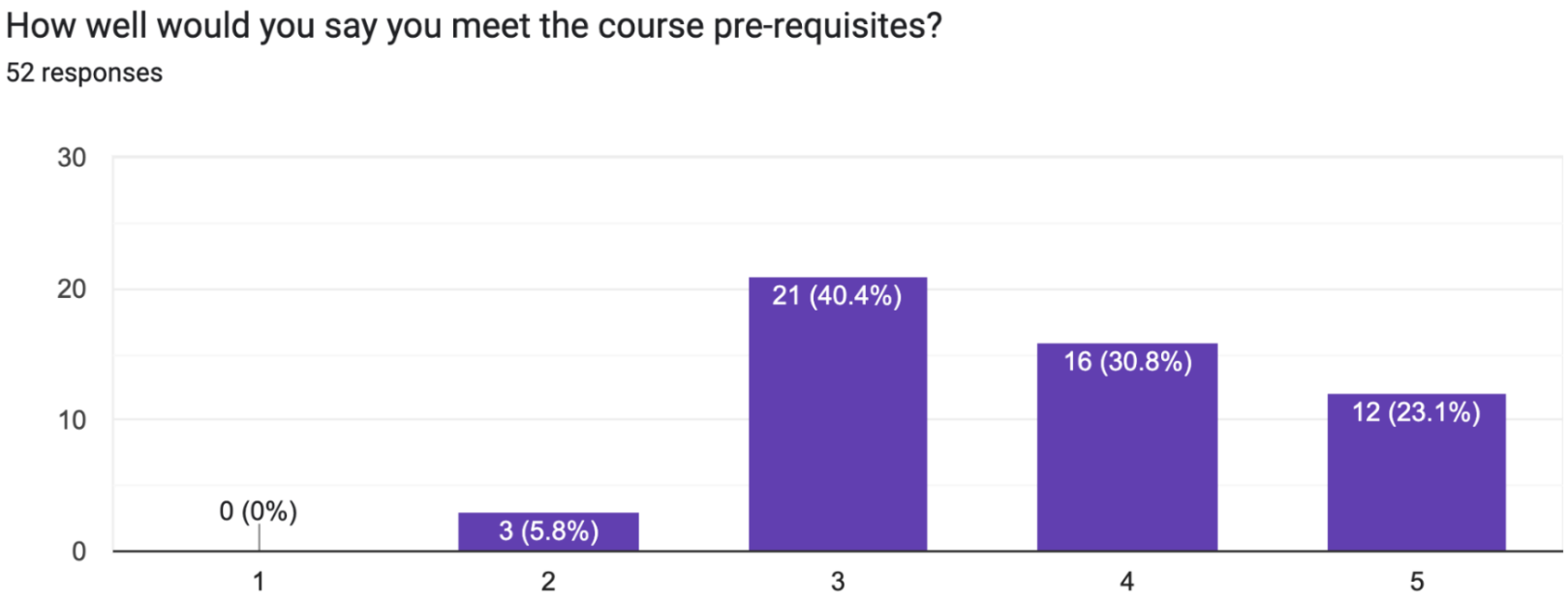
Announcements

- Midterm is almost done being graded.
- Homework 2 has been graded.
- Homework 3 is up.
  - Due next Monday.
- Final projects due in 2 weeks.
- Final exam is on August 12 (7 PM - 9 PM)
  - Please send Piazza messages for accommodation requests.
- Survey is completed (90% participation)

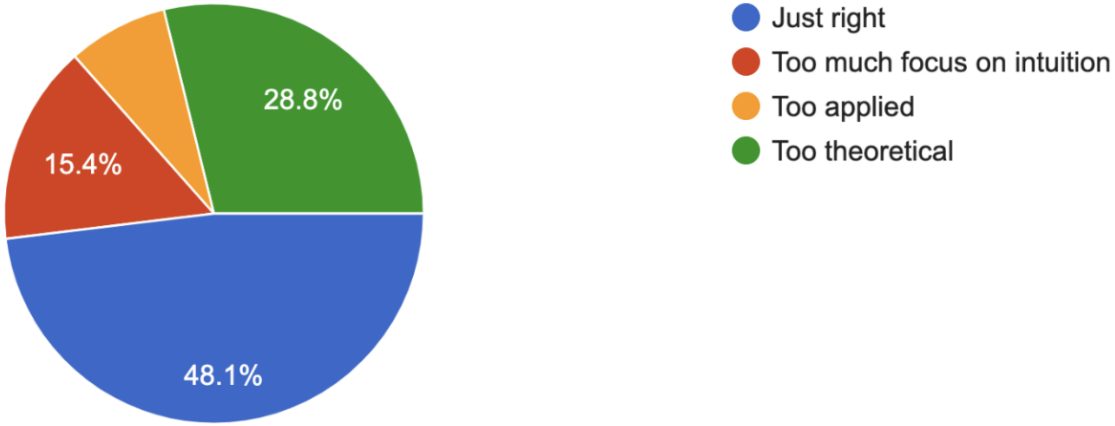
Survey
Course pre-requisites



Survey
Course material

How do you feel about the combination of course material overall (including lectures, homework, and exams)?

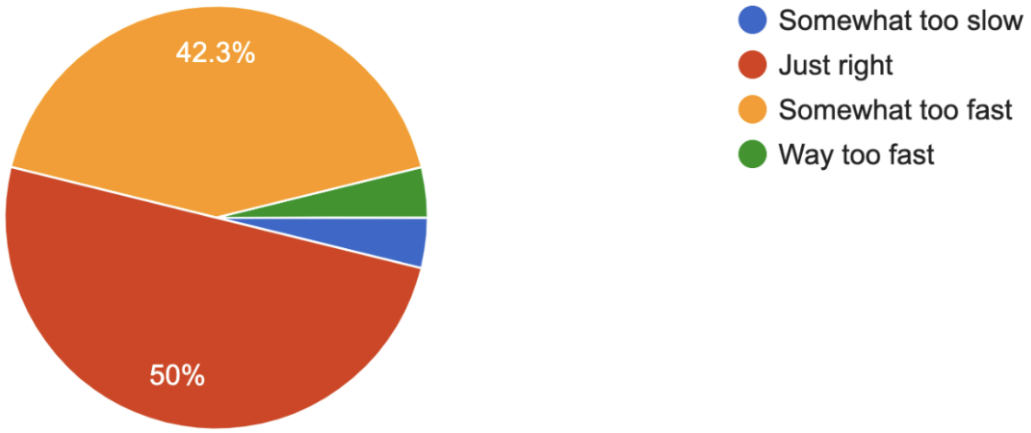
52 responses



Survey
Course pace

How do you feel about the course pace?

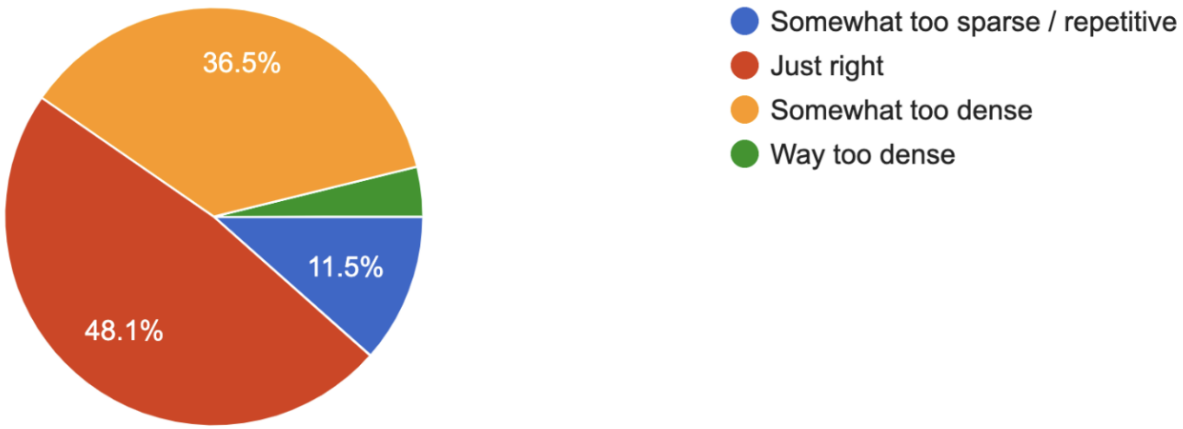
52 responses



Survey
Course density

How do you feel about the material density?

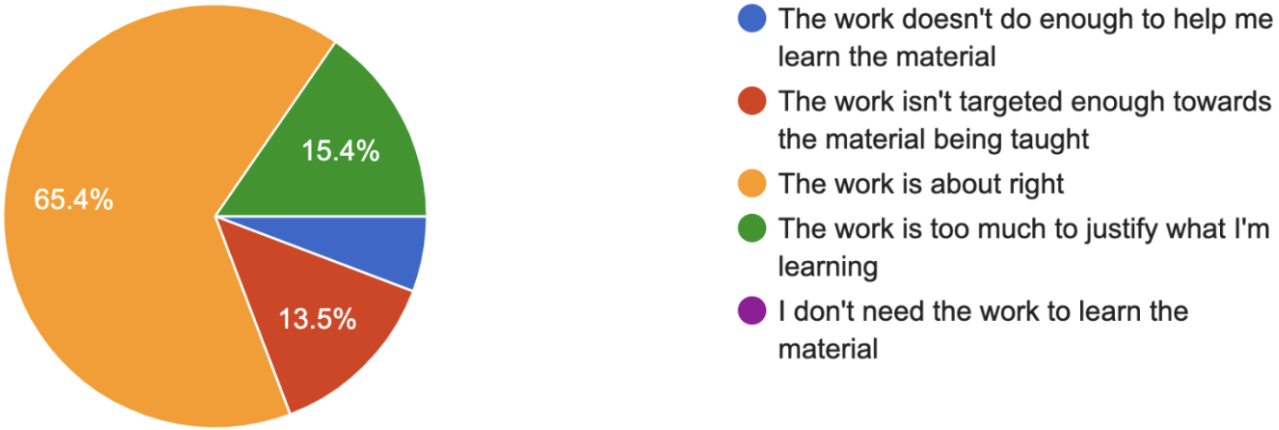
52 responses



Survey
Course workload

How do you feel about the course workload?

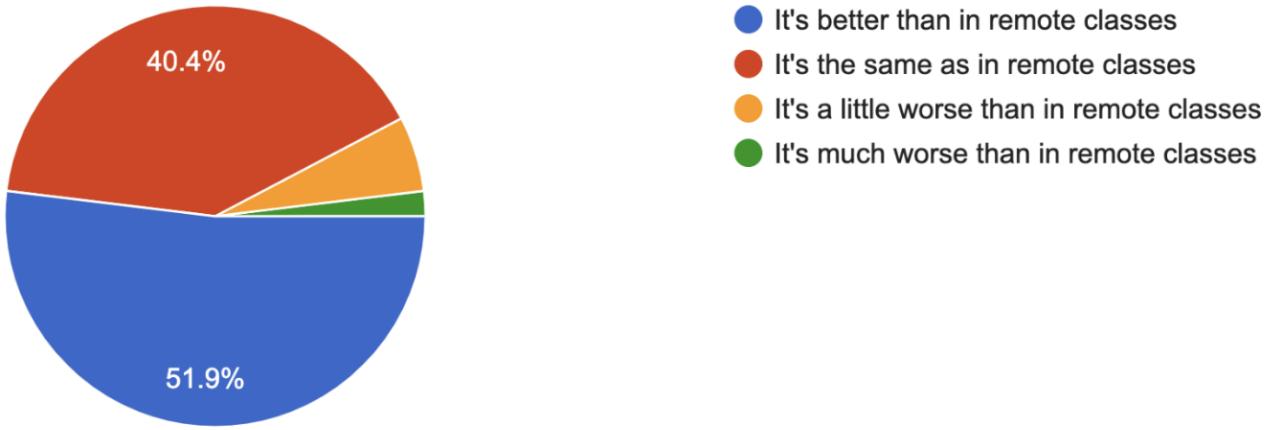
52 responses



Survey
Course in-person

How do you feel about completing the course in-person?

52 responses



Survey
Course feedback (paraphrased)

- Great course. Thank you TAs.
- We prefer in-person office hours.
- The midterm was harder than expected.
- The lectures are/aren't built enough around textbook and should/shouldn't go into more detail/derivations.

Survey
Last week

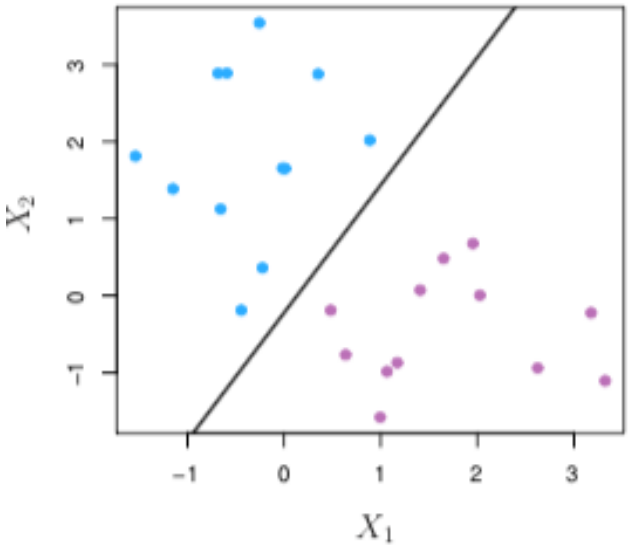
- Final project / open questions.
- Course review.
- Career / industry applications.
- Advanced methods (ex. neural networks, NLP, interpretability, heavy math/stats, etc).

Outline

Support Vector Machines

- Maximal margin classifier
- Support vector classifier
- Support vector machine

Support Vector Machines



Support vector machines are (generally) classifiers

- Linear (like logistic regression)
- Non-probabilistic (unlike logistic regression)

Hyperplanes and normal vectors

Consider a p-dimensional space of predictors

- A *hyperplane* is an affine space which separates the space into two regions

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Hyperplanes and normal vectors

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- The deviation between a point $(x_1, \ldots, x_p)$ and the hyperplane is the dot product
  - If the hyperplane goes through the origin

$$x \cdot \beta = x_1\beta_1 + \cdots + x_p\beta_p \tag{1}$$

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  - If the hyperplane is displaced from the origin by $-\beta_0$

$$\beta_0 + x \cdot \beta = \beta_0 + x_1\beta_1 + \cdots + x_p\beta_p \tag{2}$$

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$$\beta_0 + x \cdot \beta = \beta_0 + x_1\beta_1 + \cdots + x_p\beta_p \tag{2}$$

- The sign of the dot product tells us on which side of the hyperplane the point lies

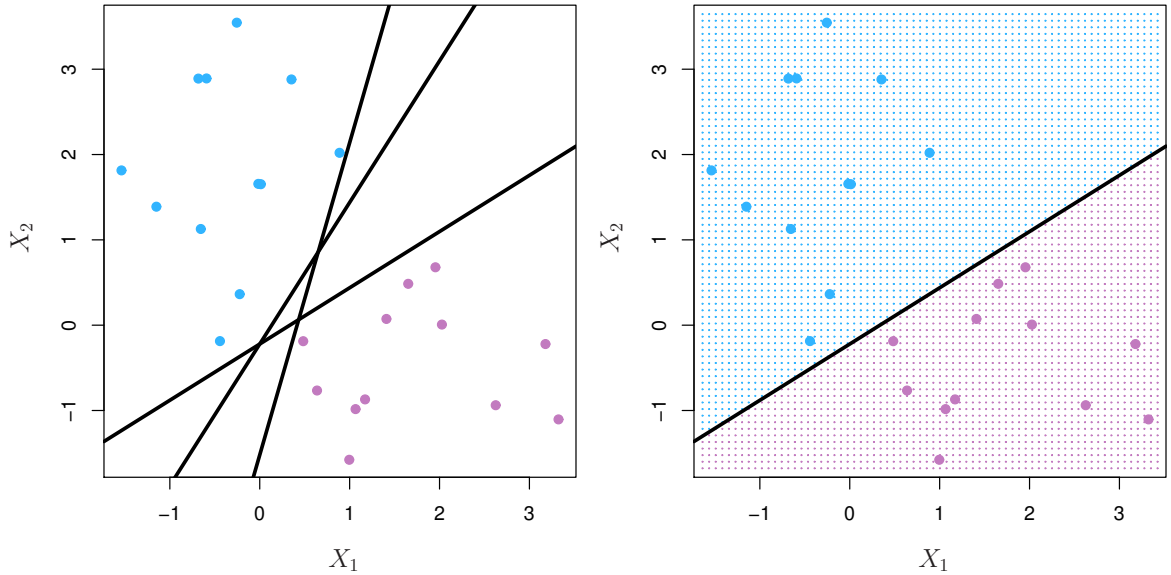
The maximal margin classifier

Suppose we have a classification problem with response $Y = -1$ or $Y = 1$.

The maximal margin classifier

Suppose we have a classification problem with response $Y = -1$ or $Y = 1$.

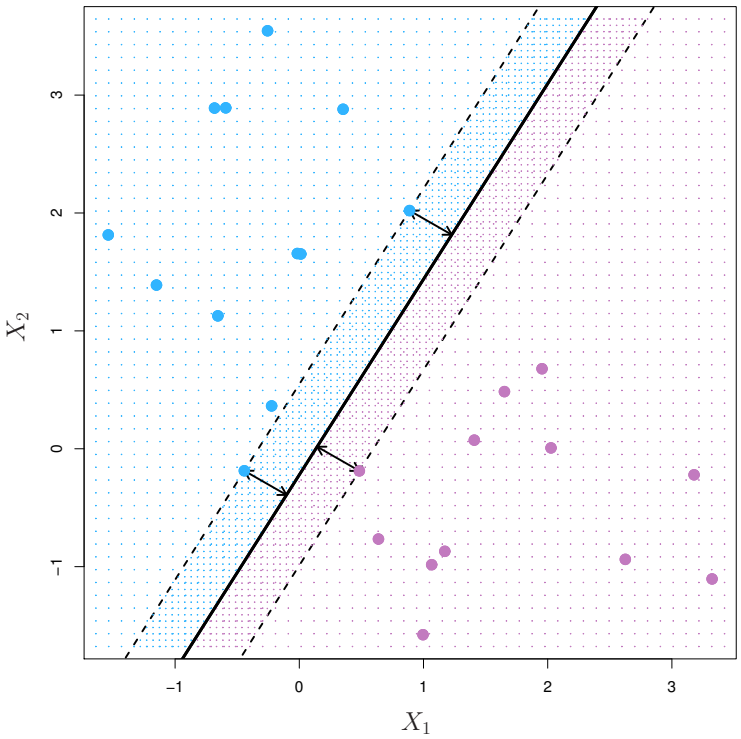
- If the classes can be separated (most likely) there will be an infinite number of hyperplanes separating the classes
- Which one should we choose?



The maximal margin classifier

Idea: Select the classifier with the *maximal margin*

- Draw the largest possible empty margin around the hyperplane
- Out of all possible hyperplanes that separate the 2 classes, choose the one with the widest margin (in both directions)



The maximal margin classifier

We can frame this as an optimization problem, i.e.

$$\max_{\beta_0, \beta_1, \ldots, \beta_p} M \tag{3}$$

subject to

- $||\beta|| = 1$
- $y_i \underbrace{(\beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip})}_{\text{How far } x_i \text{ is from the hyperplane}} \geq M \; \forall \; i = 1, \ldots, n$

where M is the width of the margin (in either direction)

n.b. the sign of $\beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}$ indicates the class

This is numerically hard to optimize

Estimating the maximal margin classifier

We can reformulate the problem by *normalizing* for $\|\beta\|$, i.e.

$$\max_{\beta_0, \beta} M \tag{4}$$

subject to

- $\frac{1}{\|\beta\|} y_i(\beta_0 + x_i^\top \beta) \geq M \; \forall \; i = 1, \ldots, n$

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or, equivalently,

- $y_i(\beta_0 + x_i^\top \beta) \geq M\|\beta\| \; \forall \; i = 1, \ldots, n$

Estimating the maximal margin classifier

Setting $||\beta|| = 1/M$, we have

$$\max_{\beta_0,\beta} \frac{1}{\|\beta\|} = \min_{\beta_0,\beta} \|\beta\| = \min_{\beta_0,\beta} \frac{1}{2}\|\beta\|^2 \tag{5}$$

subject to

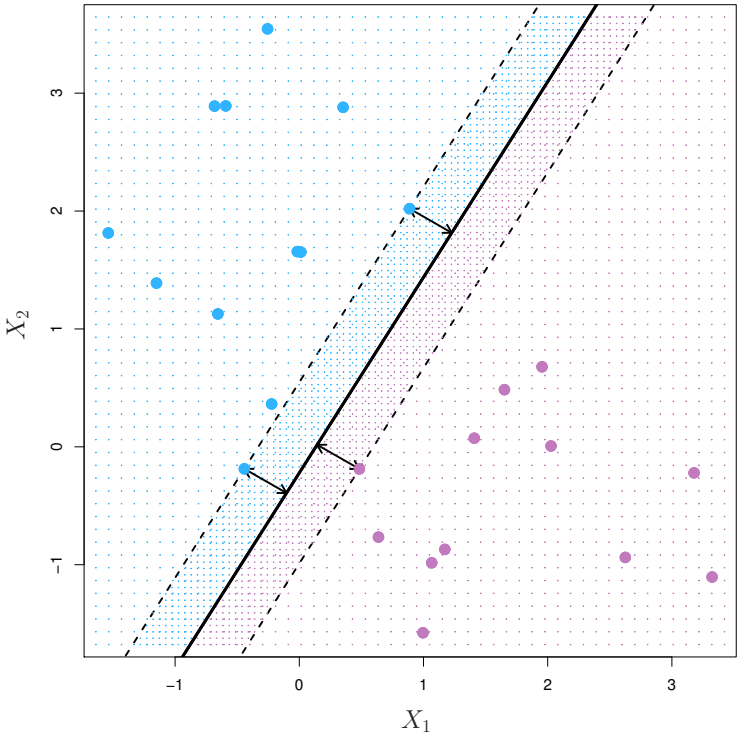
- $y_i(\beta_0 + x_i^\top \beta) \geq 1 \; \forall \; i = 1, \ldots, n$

This is a quadratic optimization problem (i.e. easier to solve)

- Typically solved using Lagrange duality

Support vectors

The vectors (i.e. observations) that fall on the margin (and determine the solution) are called *support vectors*:



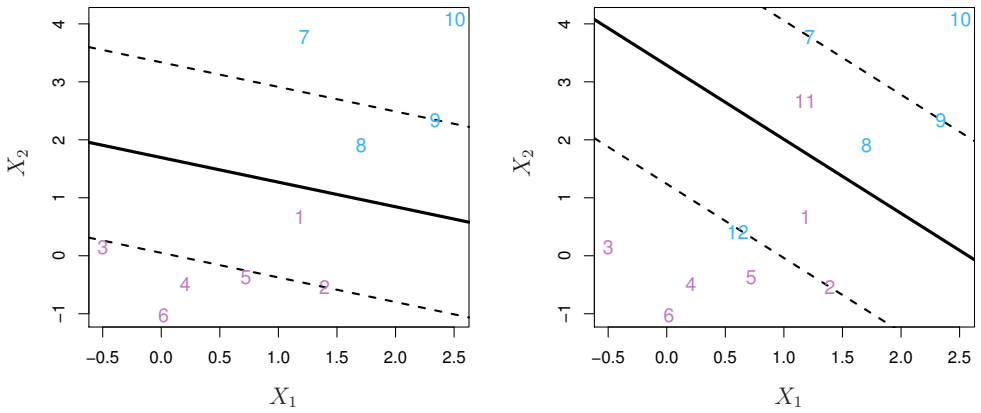
n.b. Only these points affect our estimation of the separating hyperplane.

The support vector classifier

Problem: It is not always possible (or desireable) to separate the points using a hyperplane.

Support vector classifier:

- Relaxes the maximal margin classifier, using a *soft margin*
- Allows a number of points points to be on the wrong side of the margin or hyperplane



The support vector classifier

Building this into our optimization problem gives:

$$\max_{\beta_0, \beta, \epsilon} M \tag{6}$$

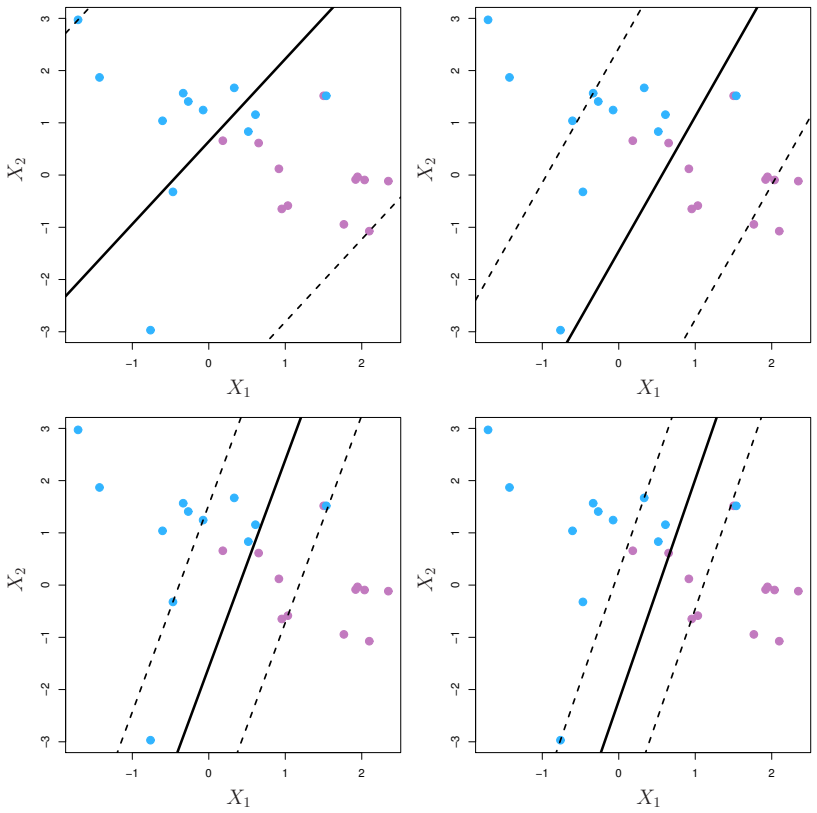
subject to

- $\|\beta\| = 1$
- $y_i(\beta_0 + x_i^\top \beta) \geq M(1 - \epsilon_i) \; \forall \; i = 1, \ldots, n$
- $\epsilon_i \geq 0 \; \forall \; i = 1, \ldots, n$ and $\sum_{i=1}^n \epsilon_i \leq C$

where

- M is the width of the margin (in either direction)
- $\epsilon = (\epsilon_1, \ldots, \epsilon_n)$ are called *slack* variables
- C is called the *budget*

Tuning the budget, C



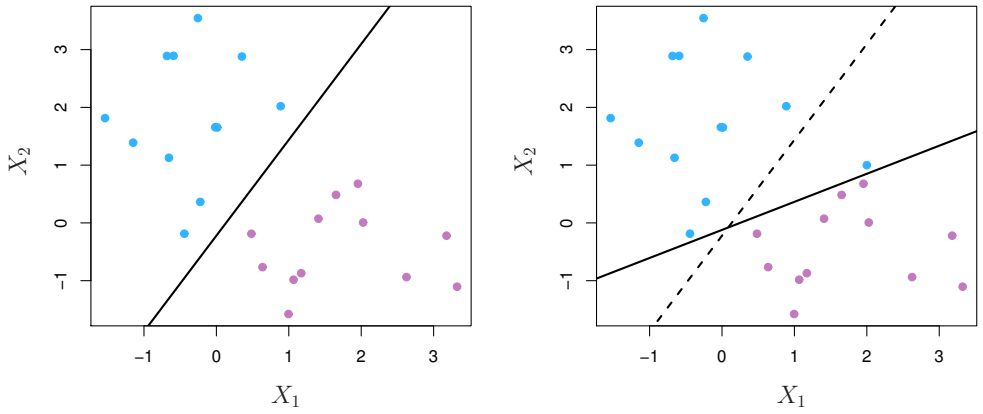
Higher C means:

- More tolerance for errors
- Larger (estimated) margins

Bias-variance trade off

C is typically chosen via cross-validation

- Larger C leads to classifers that have lower variance, but potentially have higher bias
- Smaller C leads to classifiers that are highly fit to the data, which may have low bias but high variance
  - If C is too low we can overfit
  - e.g. With the maximal margin classifier ($C = 0$), adding one observation can dramatically change the classifier



Estimating the support vector classifier

(Similar to before) we can reformulate the problem, i.e.

$$\min_{\beta_0, \beta, \epsilon} \quad \frac{1}{2}\|\beta\|^2 + D\sum_{i=1}^{n} \epsilon_i \tag{7}$$

subject to

- $y_i(\beta_0 + x_i^\top \beta) \geq (1 - \epsilon_i) \; \forall \; i = 1, \ldots, n$
- $\epsilon_i \geq 0 \; \forall \; i = 1, \ldots, n$

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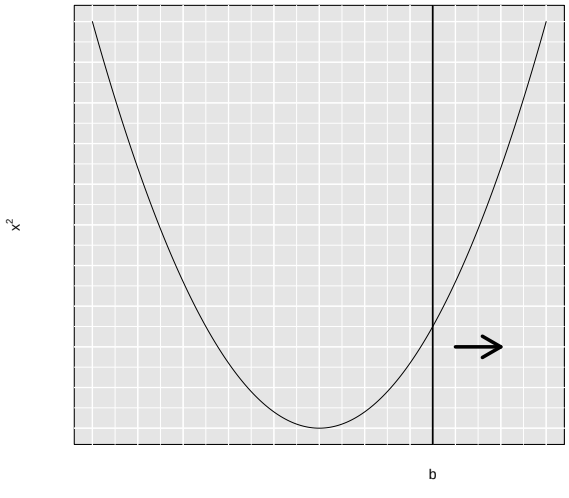
The penalty $D \geq 0$ is inversely related to C, i.e.

- Smaller D means wider (estimated) margins
- Larger D means narrower (estimated) margins

This is (still) a quadratic optimization problem

Lagrange duality

When dealing with optimization constraints



$$\min_x x^2 : x \geq b \qquad (8)$$

can be re-written as a (Lagrangian) loss function

$$L(x, \alpha) = x^2 - \alpha(x - b) \qquad (9)$$

Lagrange duality

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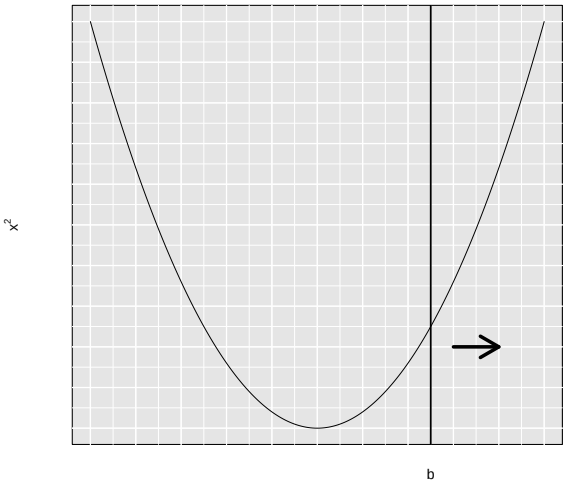

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can be re-written as a (Lagrangian) loss function

$$L(x, \alpha) = x^2 - \alpha(x - b) \qquad (9)$$

We then solve for it via

$$\min_x \max_\alpha L(x, \alpha) : \alpha \geq 0 \qquad (10)$$

Causes min to fight the max, i.e.

$$\begin{aligned} x < b &\Rightarrow \max_\alpha -\alpha(x - b) = \infty \\ x > b &\Rightarrow \max_\alpha -\alpha(x - b) = 0 \\ x = b &\Rightarrow L(x, \alpha) = x^2 - 0 \end{aligned}$$

Estimating the support vector classifier

Similar to the Maximal Margin Classifier:

- We can apply a Lagrange multipliers for our (constrained) optimization problem.
  - e.g. Karush-Kuhn-Tucker multipliers.
- This reduces our problem to finding $\alpha_1, \ldots, \alpha_n$ such that:

$$\max_{\alpha} \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i=1}^{n} \sum_{i'=1}^{n} \alpha_i \alpha_{i'} y_i y_{i'} \underbrace{(x_i \cdot x_{i'})}_{\text{inner product}} \tag{11}$$

subject to

- $0 \leq \alpha_i \leq D \; \forall \; i = 1, \ldots, n$
- $\sum_i \alpha_i y_i = 0$

This only depends on the training sample inputs through the inner products $(x_i \cdot x_j)$ for every pair of points i, j

Kernel matrix

A key property of support vector classifiers:

- To *find the hyperplane* and *make predictions* all we need is the dot product between any pair of input vectors:

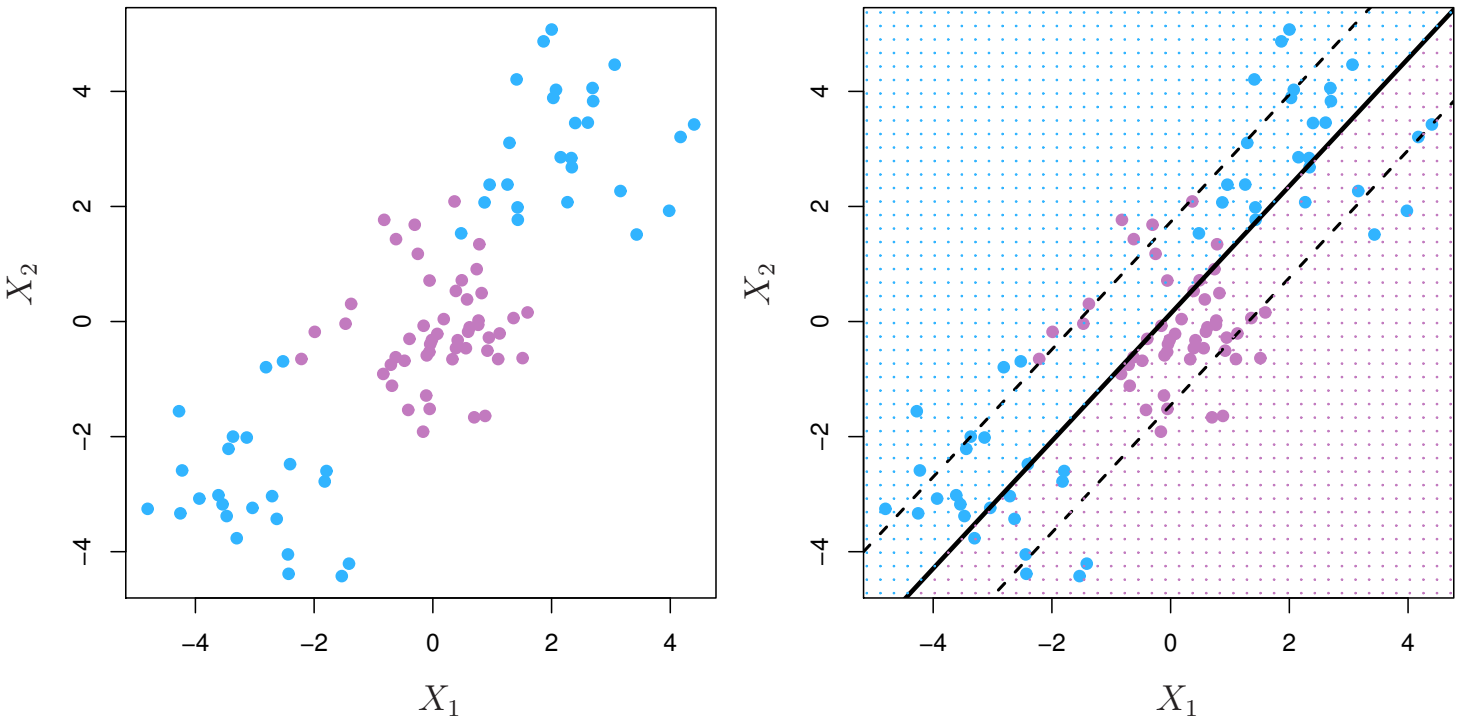
$$K(i,k) = (x_i \cdot x_k) = \langle x_i, x_k \rangle = \sum_{j=1}^{p} x_{ij} x_{kj} \tag{12}$$

- We call this the *kernel matrix*.

Non-linear boundaries

The support vector classifier can only produce a linear boundary.

Example:



Non-linear boundaries

Recall: In *logistic regression*, we dealt with this problem by adding transformations of the predictors, e.g.

- For a linear boundary:

$$\log\left[\frac{P(Y=+1|X)}{P(Y=-1|X)}\right] = \underbrace{\beta_0 + \beta_1 X_1 + \beta_2 X_2}_{\text{set equal to 0 and solve for } X_1, X_2} \tag{13}$$

- For a quadratic boundary:

$$\log\left[\frac{P(Y=+1|X)}{P(Y=-1|X)}\right] = \underbrace{\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1^2}_{\text{set equal to 0 and solve for } X_1, X_2} \tag{14}$$

Non-linear boundaries

For support vector classifiers: We can do the same thing, e.g.

- For a linear hyperplane:

$$\underbrace{\beta_0 + \beta_1 X_1 + \beta_2 X_2 = 0}_{\text{estimate } \beta\text{'s directly}} \tag{15}$$

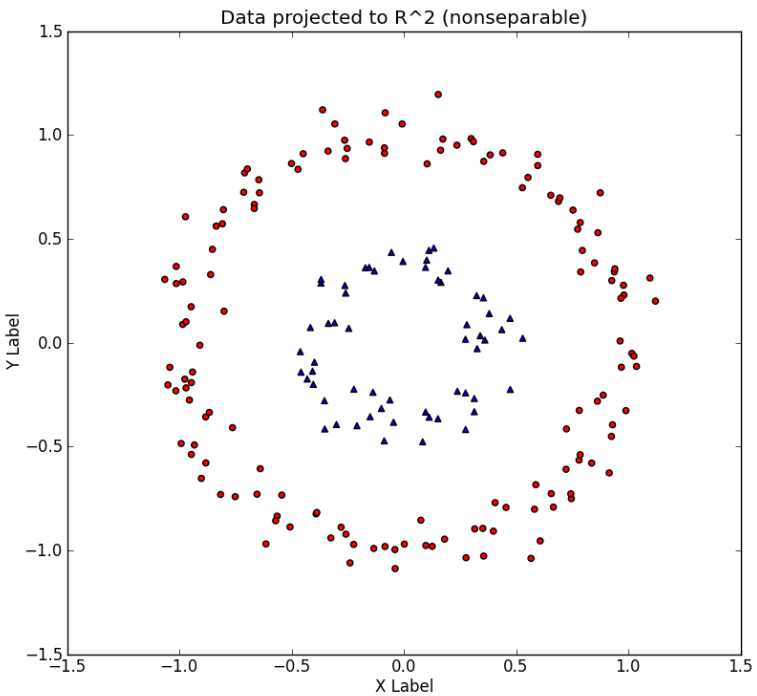
- Projecting onto a $3D$ space (X_1, X_2, X_1^2):

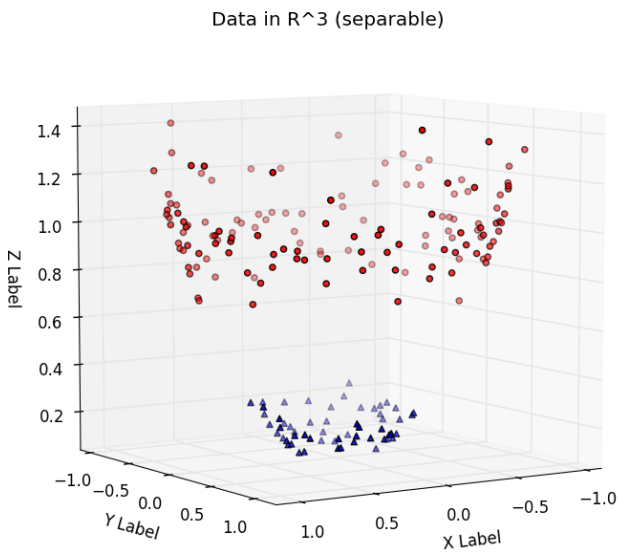
$$\underbrace{\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1^2 = 0}_{\text{estimate } \beta\text{'s directly}} \tag{16}$$

  - Still a linear boundary, but now in $3D$ space
  - Boundary is now quadratic in X_1

Non-linear boundaries

Example projection:





Left: Sample space under (X_1, X_2)
Right: Sample space under $(X_1, X_2, X_1^2 \cdot X_2^2)$

Non-linear boundaries

One approach:

- Add polynomial terms up to degree d, i.e.

$$Z = (X_1, X_1^2, \ldots, X_1^d, X_2, X_2^2, \ldots, X_2^d, \ldots, X_p, X_p^2, \ldots, X_p^d) \quad (17)$$

- Fit a support vector classifier on the expanded set of predictors

Non-linear boundaries

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Question: Does this make the computation more expensive?

Non-linear boundaries

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- Fit a support vector classifier on the expanded set of predictors

Question: Does this make the computation more expensive?

- Recall that all we need to compute is the dot product:

$$x_i \cdot x_k = \langle x_i, x_k \rangle = \sum_{j=1}^{p} x_{ij} x_{kj} \quad (18)$$

- With the expanded set of predictors, we need:

$$z_i \cdot z_k = \langle z_i, z_k \rangle = \sum_{j=1}^{p} \sum_{\ell=1}^{d} x_{ij}^{\ell} x_{kj}^{\ell} \quad (19)$$

Kernels

Rather than expanding our predictors, we could instead use *kernels* $K(i, k)$:

- Always *positive semi-definite*, i.e. it is symmetric and has no negative eigenvalues
- Quantifies the similarity of two observations

Example:
Our support vector classifier is equivalent to using the (linear) kernel

$$K(x_i, x_i') = \sum_{j=1}^{p} x_{ij} x_{i'j} \tag{20}$$

The kernel trick

Expand predictor set:

- Find a mapping Φ which expands the original set of predictors $X_1, \ldots, X_p$. For example,

$$\Phi(X) = (X_1, X_2, X_1^2)$$

- For each pair of samples, compute:

$$K(i, k) = \langle \Phi(x_i), \Phi(x_k) \rangle.$$

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Define a kernel:

- Prove that a function $f(\cdot,\cdot)$ is positive definite. For example:

$$f(x_i, x_k) = (1 + \langle x_i, x_k \rangle)^2 .$$

- For each pair of samples, compute:

$$K(i,k) = f(x_i, x_k).$$

The kernel trick

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- For each pair of samples, compute:

$$K(i,k) = f(x_i, x_k).$$

Often much easier!

The kernel trick

Example: The polynomial kernel with $d = 2$ (and $p = 2$).

$$
\begin{aligned}
K(x, x') = f(x, x') &= \big(1 + \langle x, x' \rangle\big)^2 \\
&= (1 + x_1 x_1' + x_2 x_2')^2 \\
&= 1 + 2x_1 x_1' + 2x_2 x_2' + (x_1 x_1')^2 + (x_2 x_2')^2 + 2x_1 x_1' x_2 x_2' \\
&= 1 + \sqrt{2} x_1 \sqrt{2} x_1' + \sqrt{2} x_2 \sqrt{2} x_2' + x_1^2 (x_1')^2 + x_2^2 (x_2')^2 \\
&\quad + \sqrt{2} x_1 x_2 \sqrt{2} x_1' x_2'
\end{aligned}
\tag{21}
$$

The kernel trick

Example: The polynomial kernel with $d = 2$ (and $p = 2$).

$$\begin{aligned} K(x, x') &= f(x, x') = \left(1 + \langle x, x' \rangle\right)^2 \\ &= (1 + x_1 x_1' + x_2 x_2')^2 \\ &= 1 + 2x_1 x_1' + 2x_2 x_2' + (x_1 x_1')^2 + (x_2 x_2')^2 + 2x_1 x_1' x_2 x_2' \\ &= 1 + \sqrt{2} x_1 \sqrt{2} x_1' + \sqrt{2} x_2 \sqrt{2} x_2' + x_1^2 (x_1')^2 + x_2^2 (x_2')^2 \\ &\quad + \sqrt{2} x_1 x_2 \sqrt{2} x_1' x_2' \end{aligned} \tag{21}$$

This is equivalent to the expansion:

$$\Phi(X) = (1, \sqrt{2} x_1, \sqrt{2} x_2, x_1^2, x_2^2, \sqrt{2} x_1 x_2)$$

giving us

$$K(i, k) = \langle \Phi(x_i), \Phi(x_k) \rangle \tag{22}$$

The kernel trick

Example: The polynomial kernel with $d = 2$ (and $p = 2$).

$$\begin{aligned} K(x, x') = f(x, x') &= \left(1 + \langle x, x' \rangle\right)^2 \\ &= (1 + x_1 x_1' + x_2 x_2')^2 \\ &= 1 + 2x_1 x_1' + 2x_2 x_2' + (x_1 x_1')^2 + (x_2 x_2')^2 + 2x_1 x_1' x_2 x_2' \\ &= 1 + \sqrt{2} x_1 \sqrt{2} x_1' + \sqrt{2} x_2 \sqrt{2} x_2' + x_1^2 (x_1')^2 + x_2^2 (x_2')^2 \\ &\quad + \sqrt{2} x_1 x_2 \sqrt{2} x_1' x_2' \end{aligned} \tag{21}$$

This is equivalent to the expansion:

$$\Phi(X) = (1, \sqrt{2} x_1, \sqrt{2} x_2, x_1^2, x_2^2, \sqrt{2} x_1 x_2)$$

giving us

$$K(i, k) = \langle \Phi(x_i), \Phi(x_k) \rangle \tag{22}$$

- Feature engineering is *"automated"* for us.
- Computing $K(x_i, x_k)$ directly is $O(p)$.

Defining kernels

How do we define kernels to use?

- Derive a bilinear function $f(\cdot, \cdot)$.
- Prove that $f(\cdot, \cdot)$ is positive definite (PD).

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The common approach

- Combining PD kernels we are already familiar with.
  - e.g. sums, products, etc.
- Functions of PD kernels are PD.

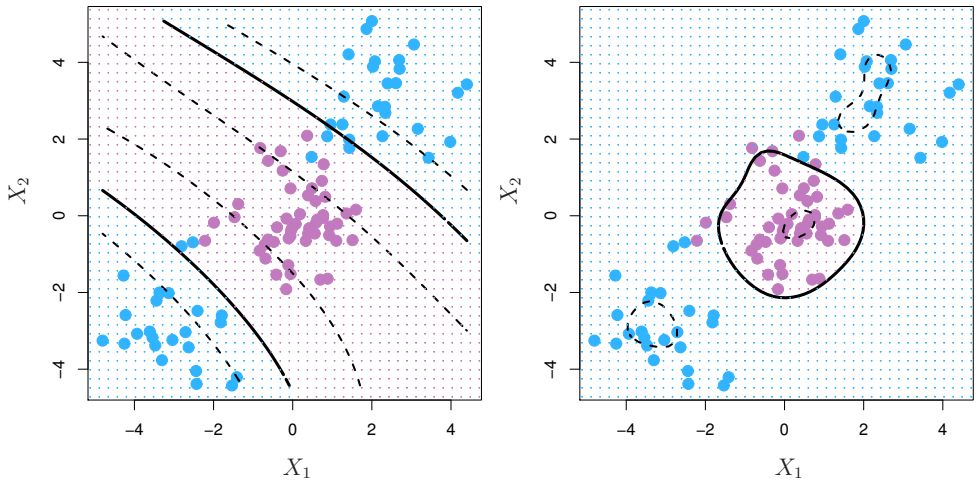
Common kernels

- The polynomial kernel:

$$K(x_i, x_k) = (1 + \langle x_i, x_k \rangle)^d$$

- The radial basis kernel:

$$K(x_i, x_k) = \exp\Big(-\gamma \underbrace{\sum_{j=1}^{p} (x_{ip} - x_{kp})^2}_{\text{Euclidean } d(x_i, x_k)} \Big)$$



Kernel properties

- Kernels define *similarity* between two samples, x_i and x_k.
- We can apply kernels even if we don't know what the transformations are.
- Kernels can result expansions that are an infinite number of transformations

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Example: Assume $p = 1$ and $\gamma > 0$

$$e^{-\gamma(x_i - x_k)^2} = e^{-\gamma x_i^2 + 2\gamma x_i x_k - \gamma x_k^2}$$

$$= e^{-\gamma x_i^2 - \gamma x_k^2}\left(1 + \frac{2\gamma x_i x_k}{1!} + \frac{(2\gamma x_i x_k)^2}{2!} + \cdots\right)$$

$$= e^{-\gamma x_i^2 - \gamma x_k^2}\left(1 \cdot 1 + \sqrt{\frac{2\gamma}{1!}} x_i \sqrt{\frac{2\gamma}{1!}} x_k + \sqrt{\frac{(2\gamma)^2}{2!}} x_i^2 \sqrt{\frac{(2\gamma)^2}{2!}} x_k^2 + \cdots\right)$$

$$= \langle \Phi(x_i), \Phi(x_k) \rangle \tag{23}$$

$$\text{where } \Phi(x) = e^{-\gamma x^2}\left[1, \sqrt{\frac{2\gamma}{1!}} x, \sqrt{\frac{(2\gamma)^2}{2!}} x^2, \cdots\right] \tag{24}$$

Multiclass approaches

SVM's don't generalize well to more than 2 class.

Two main approaches:

1. **One vs one**: Construct $\binom{k}{2}$ SVMs comparing every pair of classes. Apply all SVMs to a test observation and pick the class that wins the most one-on-one challenges.
2. **One vs all**: For each class k, construct an SVM β^k coding class k as 1 and all other classes as -1. Assign a test observation to class k^*, such that the distance from x_i to the hyperplace defined by β^k is the largest.

Relationship to logistic regression

In support vector classifiers: We can formulate

$$f(X) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p \tag{25}$$

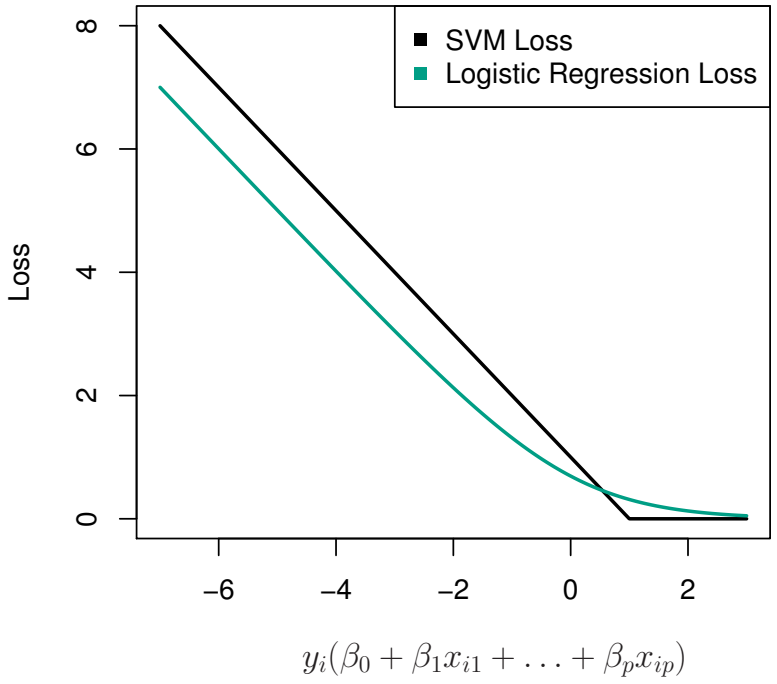
as a Loss + Penalty optimization:

$$\min_{\beta_0, \beta} \sum_{i=1}^{n} \max[0, 1 - y_i f(x_i)] + \lambda \sum_{j=1}^{p} \beta_j^2 \tag{26}$$

In logistic regression: we optimize

$$\min_{\beta_0, \beta} \sum_{i=1}^{n} \log[1 + e^{-y_i f(x_i)}] \tag{27}$$

Comparing the losses



- When the classes are well separated, SVMs behave better
- When lots of overlap in classes, logistic regression preferred

What about the kernels?

Many previous papers indicated that kernels are unique to SVMs.

- Can logistic regression also use kernels?

What about the kernels?

Many previous papers indicated that kernels are unique to SVMs.

- Can logistic regression also use kernels?

Answer: Yes (using the *Representer theorem*)

Kernel logistic regression

$$\hat{f}(x) = \log\left[\frac{\hat{P}(Y=+1|X)}{\hat{P}(Y=-1|X)}\right] \tag{28}$$

$$= \hat{\beta}_0 + \sum_{i=1}^{n} \hat{\alpha}_i K(x, x_i) \tag{29}$$

What about probabilities?

Recall: logistic regression can provide probability estimates

- Can SVMs as well?

What about probabilities?

Recall: logistic regression can provide probability estimates

- Can SVMs as well?

Answer: Yes (using logistic regression)

Let $g(x) = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}$

Platt scaling

$$\mathbb{P}(y = 1|x) = \frac{1}{1 + exp(Ag(x) + B)} \tag{30}$$

n.b. Typically done via cross-validation.

This is called Platt scaling.

References

[1] ISL. Chapter 9

[2] ESL. Chapter 12.1-12.3